

Supplementary Computational Notes for the Goldbach–Chen Frontier

Mathias Rodríguez Castro

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This supplement develops the computational reframings summarized in Appendix B of the main paper, *The Goldbach–Chen Frontier, Experimentally: A Singular-Series Cancellation and the Finite-Range Half-Power Artifact*. These notes are not used as independent evidence for the main analytic claims. They are diagnostic reformulations of the same data and are included to document additional computations, negative probes, and algorithmic side observations. The main paper remains self-contained; “the main text” below refers to that paper.

1 Goldbach as arithmetic ground-state energy

A single reformulation subsumes the counts R_1, R_2 , the layers $\Omega(q) = k$, and the partition-function viewpoint. Assign to each decomposition $a + b = N$ ($a, b \geq 2$) the *arithmetic energy* $H(a, b) = \Omega(a) + \Omega(b)$, and define the ground-state energy and level spectrum

$$E(N) = \min_{a+b=N} H(a, b), \quad n_E(N) = \#\{a + b = N : \Omega(a) + \Omega(b) = E\}.$$

Then $E(N) = 2$ iff Goldbach holds for N ; the ground-state degeneracy is $n_2(N) = R_1(N)$, and the first excited level is $n_3(N) = 2R_2(N)$ (prime+semiprime, Chen). The spectrum (Fig. 1, left) is a hump peaked at the typical energy $\langle H \rangle \sim 2 \log \log N$, with the Goldbach ground state sitting in its low-energy tail; the gap $\langle H \rangle - E(N) \sim 2 \log \log N - 2$ diverges.

The thermodynamics is computable for all N at once, since the partition function is a self-convolution,

$$Z_N(\beta) = \sum_{a+b=N} e^{-\beta H(a,b)} = (f_\beta * f_\beta)(N), \quad f_\beta(n) = e^{-\beta \Omega(n)},$$

so one FFT per β yields Z_N , the mean energy $U_N(\beta) = -\partial_\beta \log Z_N$, and the specific heat $C_N(\beta) = \beta^2 \text{Var}_\beta(H)$. As β increases past a “melting” point $\beta^*(N)$, the system localises onto the Goldbach ground state: U_N drops from $\sim 2 \log \log N$ to 2 and $C_N(\beta)$ peaks at β^* (Fig. 1, centre; mean $\beta^* \approx 2.1$). Strikingly, this melting temperature is almost a deterministic function of the singular series: $\text{corr}(\beta^*, \log \mathfrak{S}) = -0.92$ (Fig. 1, right). Where $\mathfrak{S}(N)$ is large the ground state is highly degenerate ($R_1 \propto \mathfrak{S}$) and entropically favoured, so Goldbach order melts at a *lower* β . Thus the same singular series that governs R_1 , β_2 and the balance exponent also governs the thermodynamic stability of the Goldbach ground state—a unified statistical-mechanical picture in which Goldbach is the statement “every even N has ground-state energy 2”.

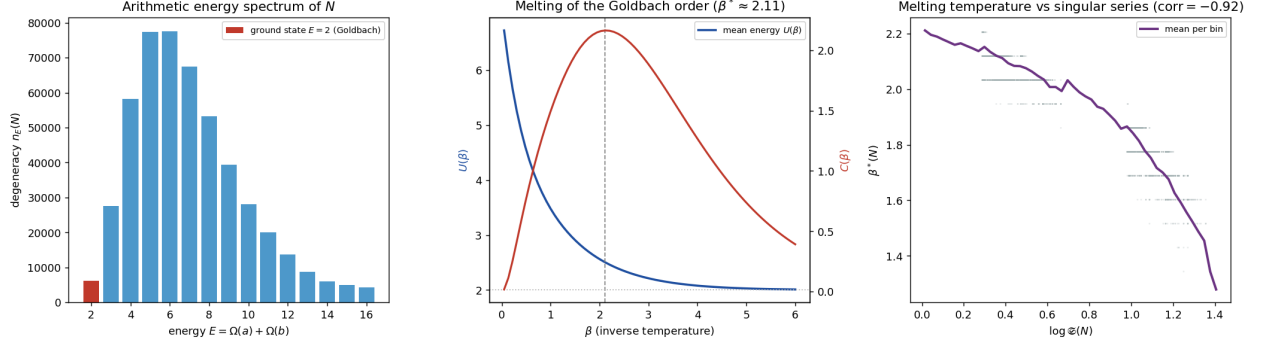


Figure 1: Arithmetic ground-state energy. Left: the energy spectrum $n_E(N)$, a hump at $\sim 2 \log \log N$ with the Goldbach ground state $E = 2$ (red) in its tail. Centre: mean energy $U(\beta) \rightarrow 2$ and specific heat $C(\beta)$, whose peak $\beta^* \approx 2.1$ is the melting temperature of Goldbach order. Right: β^* versus $\log \mathfrak{S}(N)$ (correlation -0.92).

2 Adversarial robustness of the Goldbach graph

View the representations as a bipartite graph: even integers N on one side, primes p on the other, with an edge $N \sim p$ when $N - p$ is prime; the degree of N is $R_1(N)/2$. How robust is Goldbach to deletion of primes? On a window of $1.5 \cdot 10^4$ even integers near 10^6 we compare three adversaries (Fig. 2).

Random deletion is almost powerless: deleting each prime independently with probability ϕ leaves N representable unless all its $\sim R_1(N)/2$ disjoint pairs are hit, an event of probability $(1 - (1 - \phi)^2)^{R_1(N)/2}$. Goldbach survives in the *whole* window until $\phi \approx 0.97$; one may delete 95% of all primes at random and still represent every N .

Targeted deletion is efficient but local: since the pairs $\{p, N - p\}$ of a fixed N are disjoint, breaking N is a vertex cover of a matching, costing exactly $R_1(N)/2$ primes; the cheapest target is the lowest-degree N , giving $\kappa = \min_N R_1(N)/2$. In the window this is 4.6% of the primes—an order of magnitude less than the random threshold, but aimed at a single N .

Modular deletion is the striking case. Delete the primes in one reduced class $a \bmod m$. The survivors lie in $(\mathbb{Z}/m)^\times \setminus \{a\}$, and an N is destroyed when its class is missing from the survivors' sumset. For $m = 3$, deleting the class 1 (i.e. *half* of all primes) leaves only the class 2, whose sumset is $\{2 + 2\} = \{1\}$: every $N \equiv 0, 2 \pmod{3}$ loses all its representations, destroying 59% of the window (Fig. 2, right). The same 50% of primes removed at random destroys nothing. Thus the adversary manufactures a *local obstruction*—the exact mechanism the singular series rules out for the primes themselves. The effect is confined to the smallest moduli: $m = 4$ deleting class 1 kills $N \equiv 0 \pmod{4}$ (50%), while for $m \geq 5$ the surviving classes already sum onto everything and no N is fully obstructed.

In short, the Goldbach graph is *random-robust but structure-fragile*: invulnerable to losing almost all primes by chance, yet shattered by removing one arithmetic progression. This is a network-science counterpart of the local–global theme: Goldbach's redundancy is enormous, but it is arithmetic, not statistical, redundancy.

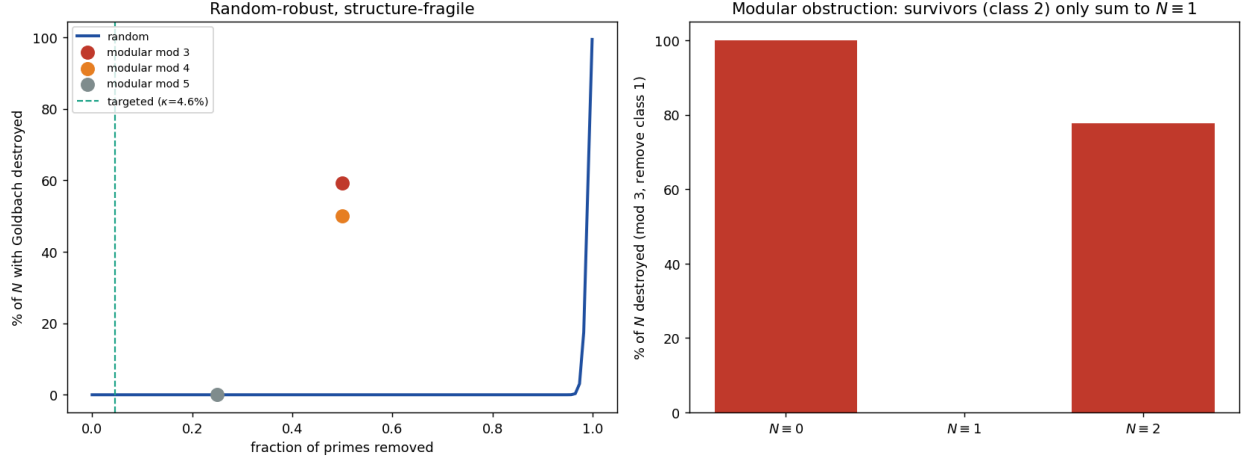


Figure 2: Robustness of the Goldbach graph. Left: fraction of window N destroyed versus fraction of primes deleted—random deletion (curve) only bites near $\phi = 1$, whereas the modular attacks at $\phi = 0.5$ (mod 3, 4) destroy 50–60%, and the targeted cut needs only $\kappa = 4.6\%$. Right: under the mod-3 attack, $N \equiv 0$ and $N \equiv 2$ are obstructed while $N \equiv 1$ is untouched.

3 A market analogy: liquidity and the fair-weather substitute

This section is a diagnostic analogy rather than an independent theorem: it asks whether the semiprime channel behaves like a substitute good under a simple prime-supply shock. Read each even N as a market that must clear by hiring two prime inputs $p + q = N$; the liquidity is the number of matches, $L_G(N) = R_1(N)$. Chen lets one input be a *minimally composite substitute* (a semiprime), giving $L_C(N) = R_1(N) + R_2(N)$ and a substitute spread $S_C(N) = R_2/R_1 \approx 2.3$: in normal times the Chen market is three times more liquid. We ask whether the substitute also provides *resilience* under a supply shock that removes each prime with probability ϕ . A Goldbach match survives with probability $(1 - \phi)^2$ (two primes), but a Chen match only with $(1 - \phi)^3$, because the semiprime $q = rs$ needs its two factors r, s to survive as well. Hence the expected surviving liquidities are $L_G(\phi) = R_1(1 - \phi)^2$ and $L_C(\phi) = L_G(\phi) + R_2(1 - \phi)^3$, and the substitute spread *decays linearly*,

$$S_C(\phi) = \frac{L_C(\phi) - L_G(\phi)}{L_G(\phi)} = S_C(0) (1 - \phi) \rightarrow 0.$$

Solving $L(\phi) = 1$ for the illiquidity threshold, both markets collapse at essentially the same point, $\phi_G \approx \phi_C \approx 0.989$, and the resilience premium $\phi_C - \phi_G \approx 2 \cdot 10^{-4}$ is negligible (Fig. 3). Within this stylised model, the economic reading is clean and a little counter-intuitive: the Chen channel is *fair-weather liquidity*—abundant when primes are plentiful, but worth ever less precisely as they become scarce, because the composite good carries a higher assembly cost. The semiprime substitute deepens the market in calm times; it does not hedge the crisis.

4 Persistent topology of additive decompositions (a negative note)

It is tempting to filter additive decompositions of N by arithmetic complexity rather than geometric distance: form the complex $K_N^{(k)} = \{\{a, b\} : a + b = N, \Omega(a) + \Omega(b) \leq k\}$ and let k grow. But for the *binary* problem this is topologically empty: each integer a lies in the single pair $\{a, N - a\}$, so $K_N^{(k)}$ is a disjoint matching for every k . Its 0-dimensional persistence is therefore exactly the energy

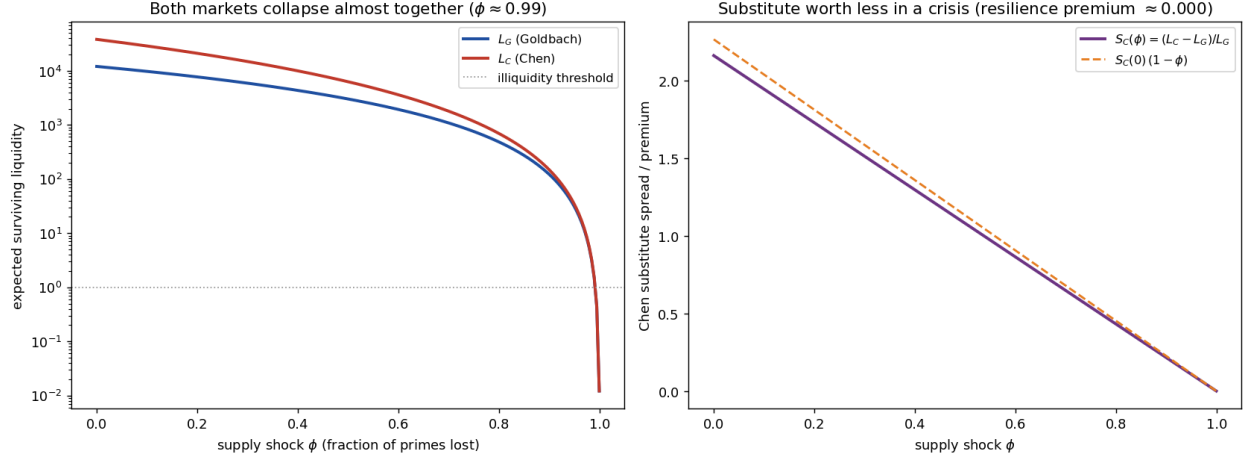


Figure 3: Goldbach as a market under a prime supply shock ϕ . Left: expected surviving liquidity of the Goldbach and Chen markets, both crossing the illiquidity threshold near $\phi = 0.99$. Right: the Chen substitute spread $S_C(\phi) = S_C(0)(1 - \phi)$ vanishes linearly—no resilience premium.

spectrum of §1—each pair is a component born at its energy $\Omega(a) + \Omega(b)$ and never dying—and all higher homology vanishes. There is no topological structure to discover beyond the spectrum.

Genuine higher dimension appears only at the *ternary* level. For odd N , form the graph on primes $p < N$ with an edge $\{p, q\}$ whenever $N - p - q$ is prime, so that each edge is a ternary representation $(p, q, N - p - q)$. At $N = 99,999$ this graph has 9,591 vertices and $3.9 \cdot 10^6$ edges (density 0.084, minimum degree 2, so it is connected—a graph form of Helfgott’s ternary theorem). Yet it is strongly *anti*-clustered: it contains far fewer triangles than a random graph of equal density, so ternary representations do not aggregate into higher-order simplices. The honest conclusion is that the additive-complexity filtration adds nothing to the spectral and graph-theoretic pictures already developed; we record it as a negative result.

5 Goldbach as control: the value of arithmetic information

Finally, view the even integers as a sequence of markets to be cleared by a controller: at each N the controller picks a prime p (the action) and clears if $q = N - p$ is prime. Goldbach is the statement that a successful policy exists; the computational question is which *simple* policy clears fastest. A naive policy ignores the arithmetic of N ; a *residue-aware* policy with information budget B uses the residues $N \bmod \ell$ for primes $\ell \leq B$ to prefer p with $N - p$ coprime to those ℓ —i.e. it avoids spending tries on partners forced to share a small factor. Over a window near 10^6 (Fig. 4), the single-shot clearing probability rises from 18% (naive) to 39% at $B = 13$, and the expected number of tries in a sequential search falls from 5.5 to 2.5—a 55% reduction. The marginal value of each prime of information is largest at $\ell = 3$ (the factor $\frac{\ell-1}{\ell-2} = 2$) and decreases thereafter, exactly tracking the singular series. Unlike the other reframings in this section, this one yields a concrete *algorithm* rather than a restatement: the residue-aware ordering is a drop-in search heuristic—enumerate candidate p in increasing order of the number of small primes $\ell \leq B$ that $N - p$ is forced to share with N —whose benchmark above (expected tries $5.5 \rightarrow 2.5$ at $B = 13$, a 55% reduction) is a measured speed-up of the per- N Goldbach search—the cost of finding one representation for a given N , distinct from the global verification $R_1 > 0$ for all $N \leq X$, which the FFT already settles—not an analogy. The same local arithmetic the singular series encodes is thus

directly actionable.

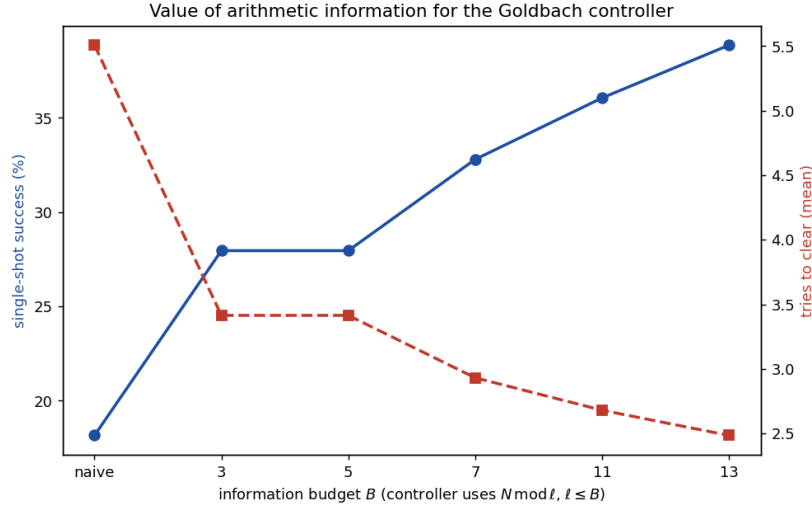


Figure 4: Value of arithmetic information for the Goldbach controller. As the information budget B grows, the single-shot clearing probability (blue) rises and the sequential clearing cost (red) falls; the gain per prime tracks the singular-series factor $(\ell - 1)/(\ell - 2)$.

6 The information content of the comet

The spectral saturation invites an information-theoretic reading: how compressible is the comet, and what is irreducible? Decomposing the variance of $\log R_1(N)$ into a smooth trend, the local arithmetic (the conditional mean given $N \bmod 30030$, which captures $\mathfrak{S}$ non-linearly), and a residual, we find (Fig. 5)

$$\underbrace{81.1\%}_{\text{trend } N/\log^2 N} + \underbrace{18.8\%}_{\text{singular series } \mathfrak{S}(N)} + \underbrace{0.11\%}_{\text{irreducible}}.$$

The comet is thus 99.9% compressible *by this particular code*—the conditional mean given $N \bmod 30030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$. We stress that this figure is relative to the chosen conditioning set, not a theorem: the “irreducible” 0.11% is what survives after removing structure up to the prime 13, and a richer conditioning (more primes, or non-residue features) would shrink it further. With that caveat, the residual is informative: its power spectrum is *flat*—white noise, in contrast to the sharply peaked spectrum of R_1 itself (the empirical circle method of the main text)—with lag-1 autocorrelation -0.04 and standard deviation 2.7%, a few times the Poisson floor $\langle R_1 \rangle^{-1/2} = 0.9\%$. So *up to modulus* 30030 the Goldbach comet is almost entirely a singular-series codeword and the residue carries no further *small-modulus* arithmetic signal; we do not claim the absence of all further structure, only that none is visible below this conditioning.

7 Discrete geometry of the Chen surface

Each Chen representation $N = p + rs$ ($q = N - p$ semiprime, $r = \text{spf}(q) \leq s$) is a lattice point on the surface $p + rs = N$. In normalised logarithmic coordinates $u = \log r / \log N$, $v = \log s / \log N$

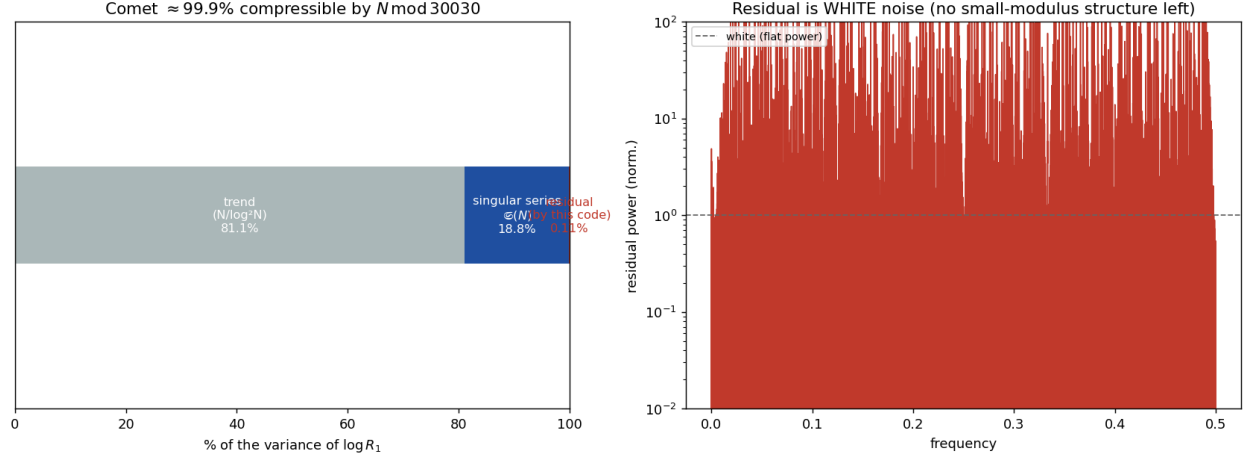


Figure 5: Information budget of the comet *relative to conditioning on $N \bmod 30030$* . Left: the variance of $\log R_1$ splits into a smooth trend (81%), the singular series (18.8%), and a residual (0.11%) that is irreducible *by this code* (richer conditioning would shrink it). Right: the residual’s power spectrum is flat—white noise, with none of the small-modulus arithmetic peaks seen in the main text’s comet spectrum.

the points fill the triangle $\{u \leq v, u + v \leq 1\}$, with the edge $u = 0$ (i.e. r bounded) the Goldbach boundary and the diagonal $u = v$ the balanced semiprimes. Accumulating $6.6 \cdot 10^7$ solutions over a window near 10^6 (Fig. 6, left) shows a cloud organised into discrete vertical *rays*—one for each small prime $r = 3, 5, 7, \dots$, since r is prime—whose brightness decreases with r : the mass piles up against the Goldbach edge.

Quantitatively, the Li–Liu exponent of a solution, $a = 1 + \log r / \log s$, has the spectrum of Fig. 6 (right): it peaks near $a \approx 1.1$ and decays toward the balanced value 2, with mean 1.38 and *median* 1.30. Half of all Chen solutions already satisfy $a < 1.3$, and essentially all sit far below the uniform threshold $a = 1.9$ of Li–Liu. So the Chen solutions of a typical N concentrate near the Goldbach boundary: the semiprime second summand is, far more often than not, an unbalanced product with a small factor—the geometric form of the residual-exponent and factor-balance findings of the main text.

8 Infinite formulation and column-generation language

This final note records the formal optimization language behind the finite selection MIPs of the main text. It is included only to document the modelling vocabulary; it is not proposed as a proof strategy. The finite selection MIPs are truncations of an infinite-dimensional feasibility problem. Let $\mathcal{E} = \{N \in \mathbb{N} : N > 2, N \equiv 0 \pmod{2}\}$; for every $N \in \mathcal{E}$, $p \in \mathbb{P}$ with $p < N$, and type $t \in \{1, 2\}$ take a binary $z_{N,p,t}$. The infinite Chen-cover is

$$\sum_{\substack{p \in \mathbb{P} \\ N-p \in \mathbb{P}}} z_{N,p,1} + \sum_{\substack{p \in \mathbb{P} \\ N-p \in \mathcal{S}_2}} z_{N,p,2} \geq 1 \quad (\forall N \in \mathcal{E}),$$

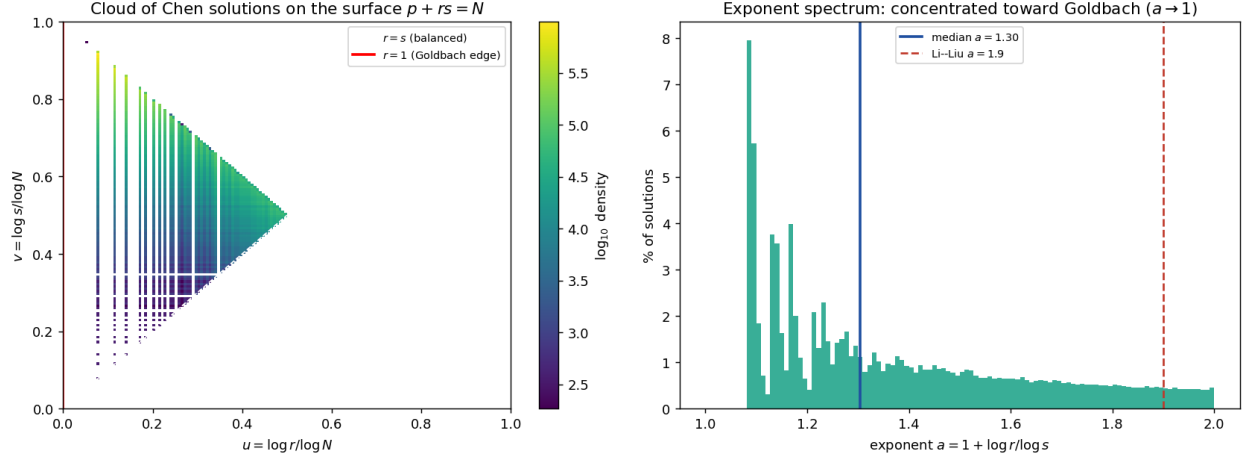


Figure 6: Discrete geometry of $p + rs = N$. Left: density of Chen solutions in $(u, v) = (\log r / \log N, \log s / \log N)$, with discrete rays at each prime r and mass against the Goldbach edge $u = 0$. Right: the exponent spectrum $a = 1 + \log r / \log s$, median 1.30, far below the Li-Liu threshold 1.9.

and its type-1 restriction is the infinite Goldbach-cover; the binary Goldbach conjecture is exactly the feasibility of the latter. One can also write a formal infinite continuity objective

$$\inf_{(p_N, q_N)} \sum_{N \in \mathcal{E}} w_N \left(\alpha |\tau(q_N) - \tau(q_{N-2})| + \beta \left| \frac{q_N}{N} - \frac{q_{N-2}}{N-2} \right| \right), \quad w_N = N^{-2},$$

a mathematically controlled version of asking whether the representation can be made continuous in the limit; finite-window truncations recover the minimum-variation MIP of the selection formulations of the main text.

Column-generation reading. The variables $z_{N,p,t}$ are columns, each a certificate that N is covered by a type- t representation: Goldbach columns (N, p, q) with $p, q \in \mathbb{P}$, Chen columns with $q \in \mathcal{P}_2$, coupled across consecutive N by the smoothness penalties. The master problem selects one column per even integer and the pricing problem seeks a representation of desirable reduced cost. As throughout, the interesting content is selection under structure—whether a low-variation path threads the representation cloud—not feasibility.